

Sets

D	set of days to schedule caregivers
C^d	set of caregivers available on day d
$C_D^d \subseteq C^d$	set of driving caregivers available on day d
L	set of localities served

Parameters

d_{ij}	distance or drive time between caregivers i and j
r_{jl}	distance or drive time between caregiver j and locality l
F_{ij}^d	number of pair visits on day d where both caregivers i and j are familiar to the client
f_{ij}^d	number of pair visits on day d where either caregiver i or j are familiar to the client
s_i^d	number of solo visits on day d where caregiver i is familiar to the client
V^d	total visit minutes on day d
V_P^d	total pair visit minutes on day d
V_s^d	total solo visit minutes on day d
$p = \frac{V_P^d}{V^d}$	share of visit minutes on day d requiring a caregiving pair
V_{Pl}^d	visit minutes in locality l on day d requiring a caregiving pair
V_{sl}^d	visit minutes in locality l on day d requiring a single caregiver
l_i	binary indicator whether caregiver i is located in locality l
K	maximum distance carers will travel for pairing

Decision Variables

v_{il}^d	variable representing the linearization of the product of binary variables $z_{il}^d y_i^d$
w_{ijl}^d	variable representing the linearization of the product of binary variables $z_{jl}^d x_{ij}^d$

$$\begin{aligned}
 x_{ij}^d &= \begin{cases} 1 & \text{if caregiver } i \in C_D^d \text{ paired with caregiver } j \in C^d \setminus \{i\} \text{ on day } d \\ 0 & \text{else} \end{cases} \\
 y_j^d &= \begin{cases} 1 & \text{if caregiver } j \in C^d \text{ is unpaired (i.e. does solo visits) on day } d \\ 0 & \text{else} \end{cases} \\
 z_{il}^d &= \begin{cases} 1 & \text{if caregiver } i \in C_D^d \text{ assigned to locality } l \in L \text{ on day } d \\ 0 & \text{else} \end{cases}
 \end{aligned}$$

$$\max \quad \sum_{d \in D} \sum_{j \in C^d} \left(s_j y_j^d + \sum_{i \in C_D^d \setminus \{j\}} \left(F_{ij}^d x_{ij}^d + f_{ij}^d x_{ij}^d \right) \right) \quad (0.1)$$

$$\min \quad \sum_{d \in D} \sum_{j \in C^d} \sum_{i \in C_D^d \setminus \{j\}} d_{ij} x_{ij}^d + \sum_{l \in L} r_{jl} \left(x_{ij}^d + y_j^d \right) \quad (0.2)$$

$$\text{s.t.} \quad y_j^d + \sum_{i \in C_D^d} x_{ij}^d = 1 \quad \forall j \in C^d \setminus C_D^d, \forall d \in D \quad (0.3)$$

$$y_j^d + \sum_{i \in C_D^d \setminus \{j\}} x_{ij}^d + \sum_{k \in C^d \setminus \{j\}} x_{jk}^d = 1 \quad \forall j \in C_D^d, \forall d \in D \quad (0.4)$$

$$\sum_{i \in C_D^d} \sum_{j \in C^d \setminus \{i\}} w_{ijl}^d \leq \lceil \frac{p|C_D^d|}{1+p} \rceil + \epsilon \quad \forall d \in D, \forall l \in L \quad (0.5)$$

$$\sum_{i \in C_D^d} \sum_{j \in C^d \setminus \{i\}} w_{ijl}^d \geq \lfloor \frac{p|C_D^d|}{1+p} \rfloor - \epsilon \quad \forall d \in D, \forall l \in L \quad (0.6)$$

$$\sum_{j \in C^d \setminus C_D^d} \left(l_j y_j^d + \sum_{i \in C_D^d} v_{il}^d \right) \leq \lceil |C_D^d| - \frac{p|C_D^d|}{1+p} \rceil + 2\epsilon \quad \forall d \in D, \forall l \in L \quad (0.7)$$

$$\sum_{j \in C^d \setminus C_D^d} \left(l_j y_j^d + \sum_{i \in C_D^d} v_{il}^d \right) \geq \lfloor |C_D^d| - \frac{p|C_D^d|}{1+p} \rfloor - 2\epsilon \quad \forall d \in D, \forall l \in L \quad (0.8)$$

$$x_{ij}^d = 0 \quad \forall i \in C_D^d, \forall j \in \{j \in C^d \mid d_{ij} \geq K\} \quad (0.9)$$

$$\lfloor \frac{p|C_D^d|}{1+p} \rfloor - \epsilon \leq \sum_{i \in C_D^d} \sum_{j \in C^d \setminus \{i\}} x_{ij}^d \leq \lceil \frac{p|C_D^d|}{1+p} \rceil + \epsilon \quad \forall d \in D \quad (0.10)$$

$$\lfloor |C_D^d| - \frac{p|C_D^d|}{1+p} \rfloor - 2\epsilon \leq \sum_{i \in C^d} y_i^d \leq \lceil |C_D^d| - \frac{p|C_D^d|}{1+p} \rceil + 2\epsilon \quad \forall d \in D \quad (0.11)$$

$$0 \leq w_{ijl}^d \leq x_{ij}^d \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \in D \quad (0.12)$$

$$w_{ijl}^d \leq z_{jl}^d \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \in D \quad (0.13)$$

$$w_{ijl}^d \geq z_{jl}^d + x_{ij}^d - 1 \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \in D \quad (0.14)$$

$$0 \leq v_{il}^d \leq y_i^d \quad \forall i \in C^d, \forall l \in L, \forall d \in D \quad (0.15)$$

$$v_{il}^d \leq z_{il}^d \quad \forall i \in C^d, \forall l \in L, \forall d \in D \quad (0.16)$$

$$v_{il}^d \geq z_{il}^d + y_i^d - 1 \quad \forall i \in C^d, \forall l \in L, \forall d \in D \quad (0.17)$$

$$x_{ij}^d \in \{0, 1\} \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall d \in D \quad (0.18)$$

$$y_i^d \in \{0, 1\} \quad \forall i \in C^d, \forall d \in D \quad (0.19)$$

$$z_{il}^d \in \{0, 1\} \quad \forall i \in C^d, \forall l \in L, \forall d \in D \quad (0.20)$$